

**SULIT**



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of  
Computing

**UNIVERSITI TEKNOLOGI MALAYSIA**  
**FINAL EXAMINATION SEMESTER II, 2017 / 2018**

**SUBJECT CODE** : **SCSR 2043**  
**SUBJECT NAME** : **OPERATING SYSTEMS**  
**SECTION** : **ALL**  
**TIME** : **9.00 AM – 11.30 AM**  
**DATE/DAY** : **01 / 06 / 2018 ( FRIDAY )**  
**VENUES** : **KTDI**

**INSTRUCTIONS :**

Answer **ALL** questions in the answer booklet given.

**(Please Write Your Lecture Name And Section In Your Answer Booklet)**

|                      |  |
|----------------------|--|
| <b>Name</b>          |  |
| <b>I/C No.</b>       |  |
| <b>Year / Course</b> |  |
| <b>Section</b>       |  |
| <b>Lecturer Name</b> |  |

This questions paper consists of **THREE ( 3 )** printed pages excluding this page.

**INSTRUCTION:** Please answer all questions in the answer booklet. Show all your workings.

1. **[20 MARKS]** There are 5 processes (in order) of *A* (115 KB), *B* (500 KB), *C* (350 KB), *D* (200 KB), and *E* (450 KB) need to be placed in the memory.
- a) Suppose a fixed-partition is used with six memory partitions of 300 KB, 600 KB, 400 KB, 200 KB, 750 KB, and 125 KB (in order).
- (i) How would the best-fit and worst-fit algorithms place the processes? Draw the memory stack and label the partition size clearly for both algorithms. [10m]
- (ii) In terms of process placement in the memory, which algorithm faced a problem? What is the problem and state the cause of the problem. [4m]
- b) Suppose a variable or dynamic partition is used. The following events happened:
- $t_1$ : process A and D left.
- $t_2$ : process F (50 KB) enters
- $t_3$ : process B left.
- $t_4$ : process G (550 KB) enters.

Draw the memory stack at each  $t_2$ ,  $t_3$  and  $t_4$  and label the sizes clearly for all processes. Use first-fit algorithm and relocatable dynamic partitions of merging adjacent holes to place the process in the memory. [6m]

2. **[10 MARKS]** Consider a logical address space of  $2^8$  pages with a 4 KB page size, mapped onto a physical memory of 4096 frames. Its page table is given in Table 2.

**Table 2**

|   |     |
|---|-----|
| 0 | 50  |
| 1 | 30  |
| 2 | 400 |
| 3 | 100 |

- a) What is the size of logical and physical addresses (in  $2^n$  bytes)? [4m]
- b) Calculate the physical address if the logical address is 3:400. [3m]
- c) What is the page number and offset for the logical address of 51152 (in decimal)? [3m]

3. [12 MARKS] Define the following terms.

- a) Demand paging [3m]
- b) Lazy swapper [3m]
- c) Belady's algorithm [3m]
- d) Thrashing in virtual memory [3m]

4. [13 MARKS] Consider the following page reference string:

**7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 1, 3**

Assuming demand paging with four frames, how many page faults would occur for the following replacement algorithms? Please circle which references cause page faults.

- a) Optimal (OPT) replacement [5m]
- b) LRU replacement [7m]

5. [10 MARKS] Draw and discuss the advantages of:

- a) Tree-structured directories [5m]
- b) Acyclic-graph directories [5m]

6. [10 MARKS] Answer the following questions:

- a) In the Unix system, directory listing is shown as below:

|            |   |     |       |      |              |         |
|------------|---|-----|-------|------|--------------|---------|
| -rw-r--r-- | 1 | pbg | staff | 9423 | Apr 19 2018  | program |
| drwx-----  | 3 | pbg | staff | 1024 | Aug 19 10:31 | mail/   |

- (i) Name and define the directory protection. [3m]
- (ii) Name and define the file protection. [3m]
- (iii) How to set all the file protection using chmod? [4m]

7. [10 MARKS] Describe and show the example of free space management using:

- a) Linked list [5m]
- b) Grouping [5m]

8. [15 MARKS] Consider the following information:

- Suppose that a disk drive has 5000 cylinders, numbered 0 – 4999.
- Using C-SCAN disk-scheduling algorithm to satisfy all the cylinder requests.
- The drive is currently serving a request at cylinder 2150.
- The queue of pending requests is:

**2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681**

Starting from current head position, answer the following questions.

- a) Draw the whole disk arm movement to satisfy all the pending requests. [7m]
- b) What is the total seek distance (in cylinder) that the disk arm had moved? [3m]
- c) Now, if C-LOOK algorithm is used to satisfy all the cylinder pending requests, compare the efficiency of both disk-scheduling algorithms based on the distance (in cylinder) that the disk arm moved. [5m]